

CURRENCY CONVERTER
COMPLETE TECHNICAL MINI PROJECT
Java-Based Currency Conversion Application

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1. Executive Summary

Currency Converter is a Java-based desktop application designed to perform currency conversion through a simple graphical user interface. The project provides currency selection, amount entry, conversion processing and a history view for previously performed conversions. The application is organized into model, service, UI and utility components, keeping data representation, conversion logic, presentation and validation responsibilities separated. The project uses Java Swing for the desktop interface. Currency-related information is maintained through the CurrencyData utility, while CurrencyService and ConversionService provide currency and conversion operations. ConversionRecord represents a completed conversion and HistoryTableModel supports presentation of conversion history. UITheme, RoundedCardPanel and CurrencyComboBoxRenderer are included to provide a customized and modern desktop interface.

2. Technology Stack

Layer	Technology	Purpose
User Interface	Java Swing	Desktop screens, controls, cards and navigation
Application	Java	Currency and conversion business logic
Model	Currency, ConversionRecord	Currency information and conversion history data
Service	CurrencyService, ConversionService	Currency lookup and conversion processing
Validation	ValidationUtil	Input validation and validation-result handling
UI Utilities	UITheme, RoundedCardPanel, CurrencyComboBoxRenderer	Theme, card styling and currency selector presentation
Execution	Java / VS Code project	Compilation and execution of the application

3. System Architecture

Java Swing UI -> Service Layer -> Model / Utility Components -> Conversion Result and History

4. Data Design

Entity / Component	Description	Key Information
Currency	Represents a supported currency	Currency identity and currency-related data
ConversionRecord	Represents a completed conversion	Source currency, target currency, amount and conversion result
CurrencyData	Supplies currency-related data	Currency information used by the application
HistoryTableModel	Prepares conversion history for the UI	Rows and columns for history display

5. Functional Modules

5.1 Currency Selection and Main Screen

The main application screen provides the controls required to select currencies, enter an amount and perform a conversion. CurrencyComboBoxRenderer is used to customize the presentation of currency selections.

5.2 Currency Conversion

ConversionService performs the conversion operation using the currency information supplied to the application. The entered amount is processed and the resulting converted value is presented to the user.

5.3 Conversion History

Completed conversions are represented through ConversionRecord objects. HistoryTableModel supports displaying these records in a tabular history view.

5.4 Input Validation

ValidationUtil provides validation support for user-entered values and returns validation information so that invalid input can be handled before conversion processing.

6. Data Validation and Reliability

Control	Implementation
Amount Validation	User-entered conversion values are validated before processing.
Currency Validation	Currency selections are handled through the application's currency components.
Conversion Result	The conversion service processes the supplied amount and selected currencies.
History Consistency	Completed conversion information is represented by ConversionRecord objects.
UI Error Handling	Validation results and application messages can be presented through the desktop interface.

7. Currency Conversion Processing Flow

1. The user opens the currency converter application.
2. The source and target currencies are selected.
3. The amount to be converted is entered.
4. The application validates the entered value.
5. The conversion service performs the required conversion.
6. The converted value is returned to the user interface.
7. The completed conversion can be represented in the conversion history.

8. Conversion and History Processing Flow

1. The user opens the conversion screen.
2. Currency selections and amount are entered.
3. The input is checked using the validation component.
4. ConversionService performs the conversion operation.
5. The result is displayed in the main interface.
6. A ConversionRecord represents the completed conversion.
7. HistoryTableModel provides the data required for history presentation.

9. Conversion Logic

Currency Conversion - The entered amount is processed by the conversion service using the selected source and target currencies.

Validation - The application validates the conversion input before attempting to process it.

Conversion Record - A completed conversion is represented as a ConversionRecord so that the operation can be retained for history presentation.

History Display - HistoryTableModel organizes conversion records for display in a table-oriented UI.

10. Build, Deployment and Configuration

The project is structured as a Java desktop application with source files organized into model, service, UI and utility packages. The archive contains Java source files as well as compiled class files. Main.java provides the application entry point, while MainFrame.java provides the principal desktop interface. The project also contains a VS Code launch configuration under the .vscode directory. No separate database configuration is present in the supplied project structure; currency information is handled by the application's Java components.

11. Conclusion

Currency Converter demonstrates a structured approach to developing a Java desktop currency conversion application using Java Swing and separated model, service, UI and utility components. The project combines currency selection, amount validation, conversion processing and conversion-history presentation in a single desktop application. The separation of responsibilities improves readability and maintainability. Components such as CurrencyService, ConversionService, ValidationUtil, ConversionRecord and HistoryTableModel provide a clear structure for handling the application's core operations. The project can be extended further with live exchange-rate services, reporting, improved history management and web or mobile integration.

12. Document Control

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